

7.1 Using AND and OR gates, find a minimum circuit to realize

$$f(a, b, c, d) = m_4 + m_6 + m_7 + m_8 + m_9 + m_{10}$$

(a) using two-level logic

(b) using three-level logic (12 gate inputs minimum)

7.2 Realize the following functions using AND and OR gates. Assume that there are no restrictions on the number of gates which can be cascaded and minimize the number of gate inputs.

(a) $AC'D + ADE' + BE' + BC' + A'D'E'$

(b) $AE + BDE + BCE + BCFG + BDFG + AFG$

7.3 Find eight different simplified two-level gate circuits to realize

$$F(a, b, c, d) = a'bd + ac'd$$

7.4 Find a minimum three-level NAND-gate circuit to realize

$$F(A, B, C, D) = \Sigma m(5, 10, 11, 12, 13) \quad (\text{four gates})$$

7.5 Realize $Z = A'D + A'C + AB'C'D'$ using four NOR gates.

7.6 Realize $Z = ABC + AD + C'D'$ using only two-input NAND gates. Use as few gates as possible.

7.7 Realize $Z = AE + BDE + BCEF$ using only two-input NOR gates. Use as few gates as possible.

7.9 Find a two-level, multiple-output AND-OR gate circuit to realize the following functions. Minimize the required number of gates (six gates minimum).

$$f_1 = ac + ad + b'd \quad \text{and} \quad f_2 = a'b' + a'd' + cd'$$

7.10 Find a minimum two-level, multiple-output AND-OR gate circuit to realize these functions.

$$f_1(a, b, c, d) = \Sigma m(3, 4, 6, 9, 11)$$

$$f_2(a, b, c, d) = \Sigma m(2, 4, 8, 10, 11, 12)$$

$$f_3(a, b, c, d) = \Sigma m(3, 6, 7, 10, 11) \quad (11 \text{ gates minimum})$$

7.11 Find a minimum two-level OR-AND circuit to simultaneously realize

$$F_1(a, b, c, d) = \Sigma m(2, 3, 8, 9, 14, 15)$$

$$F_2(a, b, c, d) = \Sigma m(0, 1, 5, 8, 9, 14, 15)$$

(minimum solution has eight gates)

7.12 Find a minimum two-level OR-AND circuit to realize the functions given in Equations (7-25) on page 215 (nine gates minimum).

$$f_1 = \Sigma m(2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

$$f_2 = \Sigma m(2, 3, 5, 6, 7, 10, 11, 14, 15)$$

$$f_3 = \Sigma m(6, 7, 8, 9, 13, 14, 15)$$

7.13 (a) Find a minimum two-level NAND-NAND circuit to realize the functions given in Equations (7-25) on page 215.

$$f_1 = \Sigma m(2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

$$f_2 = \Sigma m(2, 3, 5, 6, 7, 10, 11, 14, 15)$$

$$f_3 = \Sigma m(6, 7, 8, 9, 13, 14, 15)$$

- (b) Find a minimum two-level NOR-NOR circuit to realize the functions given in Equations (7-25).

$$f_1 = \Sigma m(2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

$$f_2 = \Sigma m(2, 3, 5, 6, 7, 10, 11, 14, 15)$$

$$f_3 = \Sigma m(6, 7, 8, 9, 13, 14, 15)$$